

The hydrogen emission spectrum

Learning outcomes

By the end of this section you should be able to describe the emission spectrum of the hydrogen atom, including the relationships between the lines and energy transitions to the first, second and third energy levels.

Hydrogen is the most abundant element – in fact it makes up approximately 90% of all the atoms in the universe. Hydrogen is located in group one of the periodic table with the alkali metals, even though it is not a metal. The reason for it being in group one is the fact that an atom of hydrogen only has one electron. In 1913, Niels Bohr proposed a model of the hydrogen atom, known as the planetary model.

- In this model the electrons were thought to travel around the nucleus in orbits (or shells), like planets orbiting the Sun.
- According to Bohr's model, only certain shells were allowed, and electrons could only exist within these shells and not between them.
- Electrons could move between the shells by either absorbing or emitting energy.

Using this model, Bohr was able to explain why hydrogen gas produced a distinctive emission line spectrum when a high voltage was passed through a sample of hydrogen gas at low pressure. However, Bohr's model only worked for atoms with one electron, namely hydrogen, but not for atoms with more than one electron.

Making connections

In [section S1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>) you learned that the electrons are located in energy levels within the atom labelled $n = 1$, $n = 2$, etc. If you need to review this section then now might be a good time to do so.

Hydrogen emission spectrum

In the previous [section S1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>), you learned that emission line spectra are produced when electrons transition from higher to lower energy levels, emitting energy in the process. In this section we will focus on the emission spectra of one element in particular, hydrogen. The [hydrogen emission spectrum](#) in the visible region, together with the electron transitions responsible for each line, is shown in **Figure 1**.

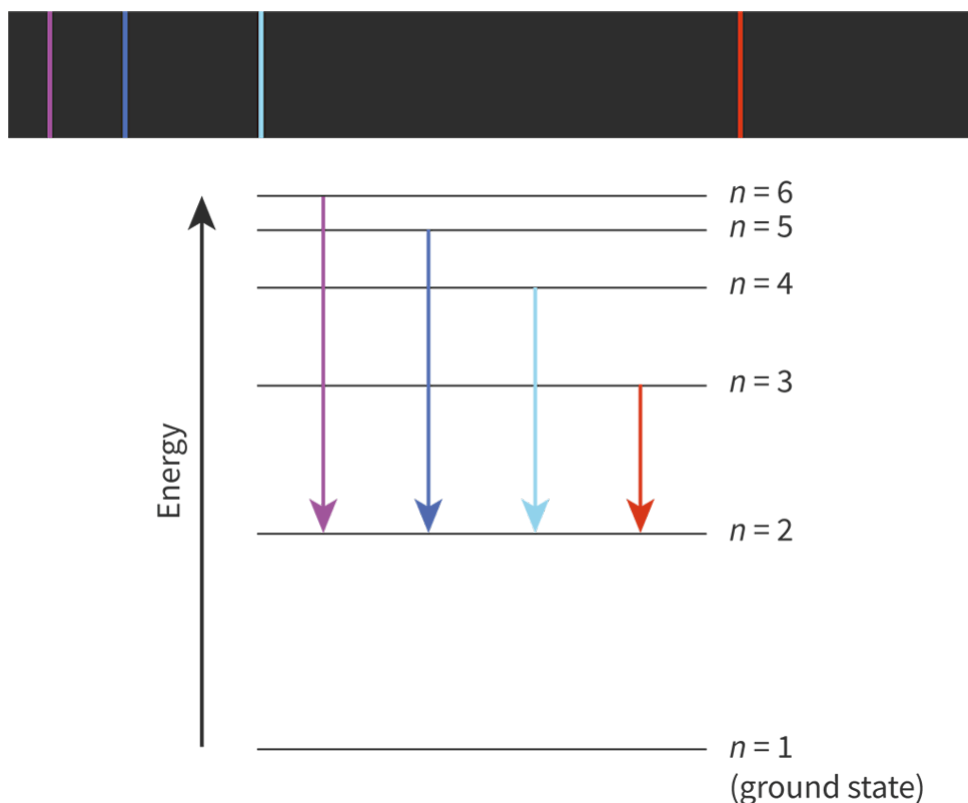


Figure 1. The hydrogen emission spectrum in the visible region together with the electrons transition responsible for the lines.

 More information for figure 1

The image shows a diagram of the hydrogen emission spectrum in the visible region along with the electron transitions responsible for each spectral line. The spectrum is depicted with discrete colored lines representing different wavelengths corresponding to electron transitions from higher to lower energy levels in a hydrogen atom. These transitions are represented with arrows between energy levels. The lines' colors range from purple to red, indicating different energy transitions, such as the transition from $n=3$ to $n=2$, which creates the red line in the spectrum, as explained in the surrounding text.

[Generated by AI]

Each electron transition corresponds to a specific line in the spectrum. For example, the electron transition from $n = 3$ to $n = 2$ produces the red line in the spectrum. Because we only see four discrete lines on the spectrum, this suggests that the electrons can only exist at certain energy levels. This provides evidence for the existence of energy levels in the atom. In other words, each line on the spectrum represents a discrete energy level in the hydrogen atom and electrons can only occupy these energy levels.

Finally, we will look at the complete hydrogen emission spectrum that shows the electron transitions from higher energy levels to the first, second and third main energy levels.

- Electron transitions from higher energy levels to $n = 1$ emit energy that corresponds to UV radiation. These represent the highest energy transitions.
- Electron transitions from higher energy levels to $n = 2$ emit energy that corresponds to visible light.
- Electron transitions from higher energy levels to $n = 3$ emit energy that corresponds to infrared radiation. These represent the lowest energy transitions.

The full hydrogen emission spectrum is shown in **Figure 2**.

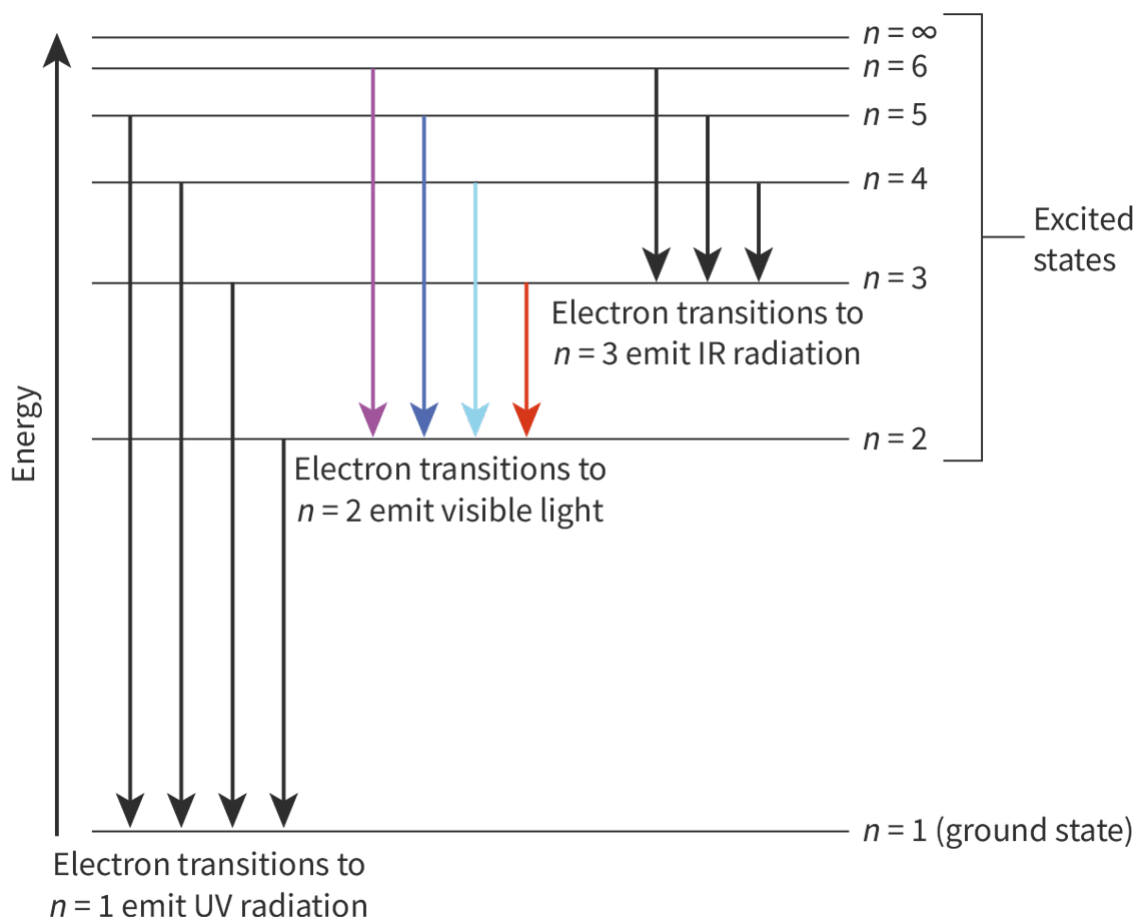


Figure 2. The hydrogen emission spectrum.

 More information for figure 2

The image is a diagram representing the hydrogen emission spectrum. It displays multiple vertical bands of color signifying different emission lines of hydrogen, such as the Lyman, Balmer, and Paschen series. Each band likely corresponds to specific wavelengths, visible as distinct color areas varied in length, representing the energy transitions within the hydrogen atom as electrons move between different energy levels. The spectrum may also suggest convergence at higher energy levels, indicating more closely spaced lines. The diagram illustrates these energy levels, possibly indicating a transition point where the electron is completely freed from the nucleus in ionization, labeled as $n = \infty$.

[Generated by AI]

A close look at the hydrogen emission spectrum reveals that the energy levels are not evenly spaced, instead they get closer together (converge) at high energies. There is also an energy level labelled as $n = \infty$. This represents the point when the electron has been completely removed from the attraction of the nucleus and the atom has been ionised.

Making connections

The point at which the energy levels converge, the convergence limit, will be used in section S1.3.6 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>) to calculate the ionisation energy (HL only).

In this activity, you will consolidate what you have learned in this section.

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

The diagram shows the energy levels of a hydrogen atom. A vertical axis on the left is labeled 'Energy' with an upward arrow. Seven horizontal lines represent discrete energy levels. Three downward arrows indicate electron transitions: one from the 5th level to the 2nd, one from the 4th level to the 3rd, and one from the 3rd level to the 1st. To the right of the levels are seven empty boxes for labeling the energy levels, and three empty boxes for labeling the transitions (A, B, and C). Below the diagram are three sets of boxes for 'Highest energy', 'Lowest energy', 'Highest frequency', 'Lowest frequency', 'Shortest wavelength', and 'Longest wavelength'. To the right of these are seven buttons labeled $n = 1$, $n = 2$, $n = 3$, $n = 4$, $n = 5$, $n = 6$, and $n = \infty$. At the bottom left is a 'Check' button with a checkmark icon.

Interactive 1. Label the Energy Levels.

More information for interactive 1

This interactive **energy level diagram** allows users to explore electron transitions in the hydrogen atom and understand their relationship with energy, frequency, and wavelength of emitted photons.

A vertical arrow on the left represents **increasing energy**. Horizontal lines represent discrete **energy levels** ($n = 1$ to $n = \infty$). The spacing between these lines **increases** as you move **down the diagram**, indicating that energy differences between lower levels are larger.

Three downward arrows show **electron transitions** from higher to lower energy levels.

Users must drag and drop the correct labels to complete the diagram:

1. Label all seven energy levels from bottom (ground state) to top (limit of ionization).
2. Label each of the three transitions with A, B, or C.
3. Order the transitions based on:

- Energy of emitted photons (highest to lowest)
- Frequency of emitted photons (highest to lowest)
- Wavelength of emitted photons (shortest to longest)

Drag-and-drop options provided are:

A

B

C

$n = 1$

$n = 2$

$n = 3$

$n = 4$

$n = 5$

$n = 6$

$n = \infty$

This interactivity helps users identify and label the energy levels and electronic transitions and understand the relationship between energy level transitions and the emitted photon's energy, frequency, and wavelength.

Read below for the solution:

Energy Levels (Bottom to Top):

- $n = 1$
- $n = 2$
- $n = 3$
- $n = 4$
- $n = 5$
- $n = 6$
- $n = \infty$

Electron Transitions:

- Arrow from $n = 4 \rightarrow n = 3$: C
- Arrow from $n = 3 \rightarrow n = 2$: B
- Arrow from $n = 2 \rightarrow n = 1$: A

Order of transitions:

Highest to lowest energy: A, B, C

Highest to lowest frequency: A, B, C

Shortest to longest wavelength: A, B, C

1. Label the energy levels on the diagram from $n = 1$ to $n = \infty$.

2. Label each arrow with A, B or C.

A represents the electron transition that emits UV radiation,

B represents the electron transition that emits visible light.

C represents the electron transition that emits IR radiation.

3. Complete the table with the energy, frequency and wavelength of the three electron transitions A, B and C.